Graphic Era HILL UNIVERSITY

Term Evaluation Theory (Even) Semester Regular Examination February 2026

Roll no. 2418159

Name of the Course: B.Tech CSE
Semester: IV
Name of the Paper: Fundamentals of Statistics and AI
Paper Code: TCS 421
Time: 1.5 hour **Maximum Marks: 50**

Note:
(i) Answer all the questions by choosing any one of the sub-questions
(ii) Each question carries 10 marks.

Q1. **(10 Marks)**

a. Explain the foundations of Artificial Intelligence. Discuss the contribution of different disciplines in the development of AI. **(CO2)**

OR

b. Describe the components of an AI program. Explain the role of each component in building an intelligent system. **(CO1)**

Q2. **(10 Marks)**

a. Describe different types of intelligent agents. Explain each type with suitable examples. **(CO1)**

OR

b. Describe problem formulation and state space representation in AI. Explain how search algorithms use these concepts. **(CO1)**

Q3. **(10 Marks)**

a. Explain Breadth First Search (BFS) algorithm in detail. Discuss its working, advantages, disadvantages, and applications. Illustrate how it is different from Depth First Search (DFS). **(CO2)**

OR

b. Describe Local Beam Search algorithm. Explain its working and advantages over simple hill climbing. **(CO1)**

Q4. **(10 Marks)**

a. Explain Depth First Iterative Deepening (DFID). Compare DFID with BFS and DFS in terms of completeness, optimality, and complexity. **(CO2)**

OR

b. Explain informed search strategies in Artificial Intelligence. How do they differ from uninformed search methods? **(CO2)**

Q5. **(10 Marks)**

a. Explain the process of problem formulation in Artificial Intelligence. Discuss the elements involved in formulating a problem. **(CO2)**

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b. Apply the **A*** search algorithm to the given state space graph. List the **OPEN and CLOSED lists** at each step and identify the final solution path. **(CO3)**

Node	h(n)
S	5
A	3
B	4
C	2
D	1
G	0

Edges:
S → A (2), S → B (1)
A → C (2)
B → D (3)
C → G (4)
D → G (2)